
Companion Review

Research Assessment, Code Audit, and Recommendations

for the repository `Daveshvats/`

`Federated-Learning-for-Distributed-Space-Weather-Monitoring` (SF-9)

1. Overall Rating: 7.0 / 10 — “Promising first application with a clear improvement path”

Dimension	Score	Rationale
Problem importance	9/10	GIC risk to grids is real, documented (1989 Québec; 2012 near-miss), and the data-sovereignty barrier is genuine.
Novelty	8/10	First FL treatment of solar flare prediction; verified against the current literature during paper preparation. Well-targeted cross-silo scenario.
Methodology	6.5/10	Sound framework (Dirichlet non-IID, FedProx, Fed-Focal, F_β thresholds), but the evaluation carries methodological debts: single seed, thresholds tuned on the test set, non-disjoint shards, and one stale metric.
Results strength	5.5/10	The direction is right (FedProx AUC 0.930 vs centralised 0.978) but absolute skill (F1 0.326, recall 0.416) is sentinel-grade, not forecaster-grade; FedAvg’s collapse is instructive but weakens the headline.
Reproducibility	7.5/10	Deterministic seed, versioned fixes (v2.1–v2.6), committed result figures; but no README, placeholder values in <code>ABSTRACT.md</code> , and metrics exist only as PNG images.
Code quality	7/10	Exceptional inline documentation and diagnostic commentary; offset by the audit findings in §3.
Overall	7.0/10	A credible workshop/short-conference contribution; the fixes in §4 would lift it toward a strong full-conference or journal paper.

2. What the research does well

1. **The right regime.** Cross-silo FL with six institution-like clients is the correct abstraction for space-weather agencies; the privacy constraint is enforced by construction, not just claimed.
2. **Honest engineering narrative.** The v2.1–v2.6 fix log (SCAFFOLD NaN root cause, focal-alpha drift, feature-schema audit) is rare, valuable documentation; the companion paper turns it into a transferable lessons section.
3. **Safety-critical framing done correctly.** Recall-weighted F_β ($\beta = 2$) threshold optimisation, rather than default 0.5-threshold metrics, matches how a grid operator would actually consume the model.

4. **Discrimination retained.** FedProx holds 95% of the centralised AUC (0.930/0.978) with zero raw-data movement — the core feasibility result stands.
5. **Interpretable, physical features.** Decoded SHAP ranks (R-value extrema, TOTUSJH, Lorentz-force variability) align with established flare physics, which strengthens scientific credibility.

3. Code Audit Findings (cross-check of the released scripts)

Ordered by severity; all were cross-checked against the committed artefacts and the extracted result values.

1. **[Bug, medium] Stale accuracy for federated models.** In `main.py`, after F_β threshold optimisation, the `accuracy` field of FedAvg/FedProx results is never recomputed (only the centralised models get `accuracy_score(...)` on new predictions). The committed `Comparison_Table.png` therefore reports FedProx accuracy as 0.019 — the value at the pre-optimisation threshold — while its confusion matrix implies ≈ 0.969 . The companion paper reports the corrected value with a footnote. *Fix:* recompute accuracy (and store predictions) for FL models exactly as done for baselines.
2. **[Bug, medium] Dirichlet shards are not disjoint.** The Dirichlet partitioner in `partition_clients.py` draws, for every (client, class) pair, a fresh random subset of the class pool with `np.random.choice`; nothing excludes indices already assigned to other clients, so training samples can appear in multiple clients (and some samples are never used). This inflates effective duplication and weakens the “six disjoint observatories” semantics. *Fix:* draw a single permutation per class and slice it by the Dirichlet proportions.
3. **[Bug, low-medium] Interpretability names lost.** The 144 statistical features are renamed `feature_0..143` in `_arrays_to_dataframe`, so the SHAP plot (and every downstream table) shows opaque indices; the companion paper had to decode them analytically. *Fix:* propagate `mean_R_VALUE`-style names end to end.
4. **[Transparency] “FedAvg” is not vanilla FedAvg.** Both `run_fedavg` and `run_fedprox` aggregate through `aggregate_weights_dafl` (distribution-aware weighting). The comparison therefore isolates the proximal term *given* DA-FL weighting — a legitimate design, but the labels in the committed figures understate this. *Fix:* either add a true-FedAvg (size-weighted only) arm, or describe DA-FL in the figure caption.
5. **[Config mismatch] LSTM vs committed outputs.** `config.py` ships `USE_LSTM=True`, but every committed artefact (FedAvg MLP, FedProx MLP) is from the MLP path with 144 statistical features. Anyone re-running “as released” will get a different pipeline than the one that produced the figures. *Fix:* pin the configuration that reproduces the committed outputs (or commit the LSTM outputs too).
6. **[Methodological debt] Threshold selected on the test set.** The F_β sweep of `main.py` runs directly on the test partitions. The optimism is bounded (one scalar from an 81-point grid) but reviewers will flag it. *Fix:* select τ^* on a carved-out validation split, report final numbers on test only.
7. **[Consistency, low] config.py BATCH_SIZE=256 is unused by the FL loop** (`_make_loader` hardcodes 512); `FOCAL_ALPHA` in `config` is bypassed by the hardcoded `focal_alpha=0.25` argument; `THRESHOLD=0.35` silently drives the misleading training-time convergence curves (the paper documents why these are uninformative). Consolidate these into one source of truth.

8. **[Hygiene, low]** ABSTRACT.md still contains [X.XX] placeholder metrics; the repo has no README (setup, data download, and run instructions live only in docstrings); results exist only as PNGs, not JSON/CSV, which blocks automated re-comparison.

4. Recommendations (prioritised)

- P0 — before any submission:** fix findings 1–2 above; add **F2** (the optimised metric!) to the reported table; run the five-seed repetition with means $\pm$ std; regenerate the figures from the fixed code.
- P1 — strengthens the science:** add a *centralised MLP* baseline (currently the federated model is compared only against a 300-tree XGBoost, which conflates architecture with privacy cost); evaluate the LSTM path; re-enable SCAFFOLD now that its SGD control-variate fix is in; replace the threshold sweep with prior correction / logit adjustment so calibration is principled rather than post hoc.
- P2 — for the venue:** a μ -sweep (10^{-3} – 10^{-1}) and α -sweep (0.5–5) to map the stability boundary; per-client evaluation (is the global model better than each client’s local model?); HARP-based geographic partitioning for realism.
- P3 — deployment track:** secure aggregation and differential privacy on the update channel (the current design protects data locality, not gradient-inversion attacks); Ditto-style personalised heads; SCADA integration notes.

5. Verdict

The repository delivers a genuinely novel, well-motivated, and honestly-engineered first study of federated solar flare prediction, with a usable feasibility result (95% of centralised AUC without data movement) and unusually good in-code documentation. Its weaknesses — single-seed evidence, sentinel-grade absolute skill, several audit-level bugs, and metrics recoverable only from images — are all fixable with bounded effort, and none undermines the core claim. With the P0/P1 items addressed, this is a solid conference paper; the companion manuscript is written to that standard so the numbers it reports will survive the fixes (the one corrected value is footnoted at its source).